WORKED SOLUTION

STEP III 1987, Question 15

Probability and Statistics • Expected Cost Optimisation

STEP III 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q15

The key idea

Rather than minimising E_n directly, look at what one *extra* test changes. Comparing the strategies "stop after n" and "stop after $n+1$" only matters for patients who have had n positive results so far. For those patients, the extra test costs £10, and it saves the £100 treatment for a Steppe-syndrome patient whose test comes back negative. So the sign of $E_{n+1} - E_n$ decides everything, and that sign changes at most once.

Solution

Let F be the event that the patient has Fenland fever, so $\mathrm{P}(F) = p$, and let $S = F'$ be Steppe syndrome. We assume the test results for a Steppe patient are independent, each positive with probability q, where $0 < q < 1$. Treatment costs £100 whoever receives it.

Part (i)

Finding E_n. If the patient has F, every test is positive. So all n tests are given and then the treatment, at a cost of $10n + 100$.

If the patient has S, test number k (for $1 \leq k \leq n$) is given only when the first $k-1$ tests were all positive, which happens with probability q^{k-1}. So the expected number of tests is

$$\sum_{k=1}^{n} q^{k-1} = \frac{1-q^n}{1-q},$$

and the patient is treated only if all n tests are positive, with probability q^n. By linearity of expectation,

$$E_n = p\,(10n+100) + (1-p)\left(\frac{10(1-q^n)}{1-q} + 100q^n\right).$$

This also gives $E_0 = 100$, as it should, since then everyone is treated.

The difference. The F part increases by 10. In the S part, $\dfrac{1-q^{n+1}}{1-q} - \dfrac{1-q^n}{1-q} = q^n$, so

$$\begin{aligned} E_{n+1} - E_n &= 10p + (1-p)(10q^n + 100q^{n+1} - 100q^n) \\ &= 10p + 10(1-p)q^n(1 + 10q - 10) \\ &= 10p - 10(1-p)q^n(9 - 10q), \end{aligned}$$

as required.

Reading the result: with probability p, test $n+1$ is wasted on a fever patient. With probability $(1-p)q^n$, it is given to a Steppe patient and costs £10, but with probability $1-q$ it saves £100. The net expected saving is $100(1-q) - 10 = 10(9 - 10q)$.

The numbers. Write $D_n = E_{n+1} - E_n$. With $p = 10^{-4}$ and $q = 10^{-2}$ we have $9 - 10q = 8.9$ and $1 - p = 0.9999$, so

$$D_n = 10^{-3} - 89 \times 0.9999 \times 10^{-2n}.$$

Ask yourself Why is it enough to find where D_n changes sign?

Here $9 - 10q > 0$ and q^n strictly decreases as n grows, so D_n strictly increases with n. Hence the sequence E_n goes down while $D_n < 0$ and up once $D_n > 0$, and the minimum is at the first n with $D_n > 0$. Now

$$\begin{aligned} D_2 &= 10^{-3} - 0.0088991 < 0, \\ D_3 &= 10^{-3} - 0.000088991 = 0.000911\ldots > 0. \end{aligned}$$

So $E_0 > E_1 > E_2 > E_3$ and $E_3 < E_4 < E_5 < \cdots$, and the doctor should choose $n = 3$. (The expected cost is then about £10.11, against £100 for treating at once.)

Part (ii)

Take $q_0 = \frac{9}{10}$, and suppose $q > \frac{9}{10}$. Then $9 - 10q < 0$, so

$$D_n = 10p + 10(1-p)q^n(10q - 9).$$

Both terms are non-negative for every $p \in [0,1]$. They can't both be zero: if $p > 0$, the first term is positive, and if $p = 0$, the second is $10q^n(10q - 9) > 0$. So $D_n > 0$ for every $n \geq 0$, which means

$$E_0 < E_1 < E_2 < \cdots.$$

Whatever the value of p, the doctor should choose $n = 0$ and treat at once.

This value of q_0 can't be lowered. If $q < \frac{9}{10}$, then $D_0 = 10p - 10(1-p)(9 - 10q)$ is negative for small enough p, so testing at least once beats $n = 0$. In words, when $q > \frac{9}{10}$ a Steppe patient's test is negative less than one time in ten, so a £10 test saves less than £10 of treatment on average, even before counting the fever patients.

Checks

- $n = 1$ directly: $E_1 = 10 + 100\big(p + (1-p)q\big)$. With $p = 0$ this gives $E_1 - E_0 = 10 - 100(1-q)$, matching the formula's $-10q^0(9 - 10q)$.
- With $p = 10^{-4}$ and $q = 10^{-2}$ the formula gives $E_0 = 100$, $E_1 \approx 11.010$, $E_2 \approx 10.1210$, $E_3 \approx 10.11309$ and $E_4 \approx 10.11400$. The minimum is at $n = 3$, as found.
- Exact enumeration with fractions confirmed the formula for E_n and the difference identity for hundreds of random (p, q) and $n \leq 8$, and a simulation with $p = 0.3$, $q = 0.5$, $n = 2$ gave a mean cost of 63.98, against the exact 64.

Common slips

- Charging a Steppe patient for all n tests. The process stops at the first negative result, so test k is given only with probability q^{k-1}.
- Finding the first n with $D_n > 0$ without saying why D_n is monotonic. Without that, the sequence E_n could in principle fall again later.
- In part (ii), forgetting the case $p = 0$. That case is why the condition is $q > q_0$ rather than $q \geq q_0$: at $q = \frac{9}{10}$ and $p = 0$, every E_n equals 100.

Beyond STEP

This is a small example of *optimal stopping* and of decision theory: a sequence of actions with costs, a random state, and a rule for when to stop. Comparing neighbouring strategies ("marginal analysis") and showing the differences change sign only once is the standard way to handle such problems. The ideas lead on to Bayesian decision theory and dynamic programming. Grimmett and Welsh, *Probability: An Introduction*, covers the conditional-expectation tools that underpin them.

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